

ModPaddle : A Racket-Shaped Force Feedback System for Modifying User's Swing Power Against Moving Ball

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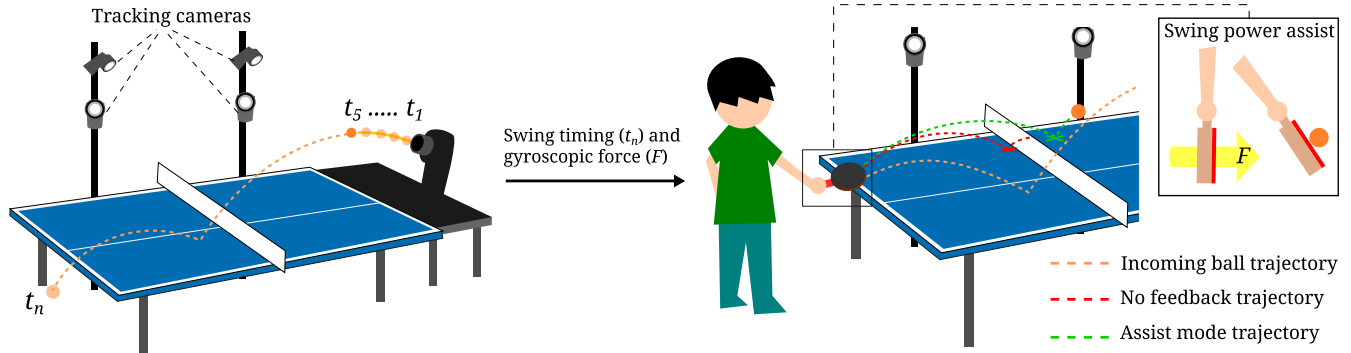


Figure 1: Illustration of the proposed force feedback system to modify the user's swing power against moving ball. Left: the system uses four tracking cameras to track the ball and estimate its trajectory, speed and swing timing (t_n). Right: the system sends swing timing and the gyroscopic force (F) to the racket. The amount of gyroscopic force can be adjusted.

Abstract

In table tennis, swing power is significantly crucial for regulating the speed and trajectory of the ball. However, traditional coaching feedback struggles to provide assistance on instructing the time to swing and the amount of power required to return the ball, especially during the user's swing motion. To tackle such a challenge, we present ModPaddle, a racket-shaped force feedback system that can modify the user's swing power and provide real time assistance. ModPaddle utilizes the gyroscopic effect to generate a force that can either assist or limit the user's swing power and provide the swing timing. The proposed system can generate gyroscopic force up to 7.56 N, enabling the user to successfully return the ball to proper range of distance on the opponent's field. ModPaddle aims to revolutionize table tennis training by providing instant, tangible guidance potentially teaching players the correct amount of power and swing timing to return the ball to desired distance.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Interaction devices.

Keywords

Table tennis, Force feedback, Gyroscopic effect

ACM Reference Format:

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1 Introduction

In sports training, verbal feedback, demonstrating correct techniques, and hands-on guidance from coaches to players are some of the conventional teaching methods. Through these conventional teaching methods, coaches' communication skills significantly affect players' skill acquisition in sports [23]. However, no matter how good coaches' communication is, each athlete may have a different interpretation, thus producing possibly different outcomes and learning times.

In table tennis, one of the common novice's mistakes is hitting the ball too hard [15, 16]. Novice and intermediate players usually try to emulate the advanced level players by hitting the ball with a lot of power. Logically, hitting the ball harder can produce a lot of speed to the ball and make it difficult for the opponent to return. While it's a good winning strategy, doing it constantly is not an easy feat. Usually, novice and intermediate players who often hit the ball too hard, miss a lot more points because the ball often flies out of the court before bouncing on the opponent's court [16]. In addition, it is not easy to give detailed instructions such as the

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timing to swing and the amount of swing power to hit the ball, especially when athletes are in the middle of swing motion and focused on the task at hand [5, 18, 24].

To facilitate learning and improve communication in sports training, some studies have proposed several sports assist systems using visual feedback [7, 10, 13, 26–28]. The visual feedback can be categorized based on the time when feedback is displayed, including delayed feedback [13, 27, 28] and immediate feedback [7, 10, 20]. In delayed feedback, the visual feedback is usually displayed after the player completes their motion to evaluate how the player performed [13, 28]. It is also possible to immediately display the feedback during the motion to guide the player to modify their motion [7, 10, 20]. While delayed feedback is useful, it's difficult to translate the feedback into optimized motion. A player's racket swing power and timing can vary with each attempt, making it challenging to provide a consistent solution for the next swing motion. In contrast, immediate feedback is more effective in improving player performance [7]. Such feedback can help modify user's racket swing during the swing motion, thus optimized swing can be achieved. However, due to humans' limited range of vision, simultaneous checking and doing the motion can be difficult [18]. For example, Wu et al. described in their study that visual cues in table tennis can distract non-athletic users as they focus more on the ball rather than the visual cues [26].

Other than visual feedback, several studies have explored the use of force feedback in sports-related devices. While there are not many force feedback devices for table tennis, there are several feedback devices that have the potential to be used in table tennis. Force feedback devices often provided the feedback immediately, allowing the device to modify user's motion. These devices can be categorized based on their force feedback generation methods, including wire-driven systems [8, 9], electrical muscle stimulation (EMS) [4, 5], air jet [24], and gyroscopic effect [18, 19].

Hirata et al. proposed a passive wire-driven tennis swing motion support system using servo brakes [8]. A tennis racket is attached to several wires connected to servo brakes. The wire tension can be controlled with servo brakes to modify user's racket position and orientation during swing. In another study, Huang et al. proposed a planar four-cable assisted golf putting system [9]. Similar to Hirata et al. the head of the putter is attached to four wires connected to the motors. The wire can control the angle of the putter's head to help users improve their putting accuracy. However, in both scenarios, the range of the user's body movement is significantly affected by the length of the wire.

Several studies introduced the use cases of EMS as a force feedback [4, 5]. Faltaous et al. proposed EMS to be attached at the user's calf [5]. The system produced force feedback in the user's calf that allowed coaches to immediately guide trainees to maintain a ready position in racket-based sports. Similarly, Faltaous et al. proposed the attachment of EMS on users' wrists for golf [4]. The system produced force feedback on the user's wrist to modify the turning angle of the wrist and guide the user to more accurate golf putting. While EMS offers a potential method for generating force feedback in sports assist, it comes with notable drawbacks. Studies indicate that users often experience a diminished sense of control over their movements when EMS is applied, resulting in reduced

sense of agency [18, 22]. Furthermore, many users prefer to maintain control over their movements instead of having their muscle stimulated.

Suzuki et al. introduced an ungrounded air jet force feedback system to modify the user's racket pose in table tennis [24]. They installed four air tubes in the front and back of the racket to produce air pressure that can rotate and move the racket. Compared with the device proposed by Hirata et al. [8], the user has more freedom when using this device. However, the possibility of interference by air pressure coming out of the racket should be considered when the real ball is played.

Another approach to generate force feedback is by utilizing the gyroscopic effect. Nakamura et al. proposed the use case of the gyroscopic effect for golf [18, 19]. Their system consists of two rotating weights (each connected to a motor) and a pan-tilt mechanism to generate torque as force feedback. Their system is combined with motion capture to estimate if the user's swing trajectory deviates from the ideal trajectory. When the user's swing deviates, the golf club generates force feedback through a gyroscopic moment to correct the user's golf club swing. While their system is interesting, it is only considered in sports where the ball doesn't move.

In this study, we present ModPaddle, a racket-shaped force feedback device for adjusting the user's racket swing power and provide swing timing against moving ball. The system utilizes the gyroscopic effect which to the authors' knowledge previously has not been implemented on sports where the ball is constantly moving such as table tennis. Our proposed system provides instant feedback during the user's swing motion. A range of force can be produced by the proposed racket enabling sports coaches to directly instruct the appropriate amount of the swing power that the player needs.

2 System Design

2.1 Gyroscopic Force Feedback Device

We proposed a racket-shaped force feedback device for table tennis. Figure 2 shows the design and assembled prototype of our racket device. The device consists of two DC motors (DC motor A: T-Motor MN5006 KV450, DC motor B: Pololu-3716 20D Metal Gearmotors), an electronic speed controller (Beatles G2 20A), a 114 mm diameter flywheel, a microcontroller (Raspberry Pi Zero 2W), and a DC power supply. The device is assembled with a handle and a case to make the shape of a table tennis racket. The flywheel, handle, and case were produced with a 3D printer (Raise3D Pro3 Plus) using PLA. The racket blade is produced with 2 mm thick wood and standard table tennis rubber on one side and 2 mm thick acrylic on the other. We designed the case to be 65 mm thick to accommodate the movement of the gyroscope. The dimension of the case and the blade is set to be close to real table tennis racket of 172 mm × 145 mm. We add several holes to the case to reduce the total weight of the racket, resulting in an assembled weight of approximately 530 g. The racket is powered with a DC power supply with an input voltage of 20 V.

We also measured and compared the center of mass of the proposed racket with that of a standard table tennis racket. The center of mass of the proposed racket and the standard table tennis racket were located at approximately 12 cm (closer to the handle) and 10 cm (almost at the center of the blade), respectively, from the tip

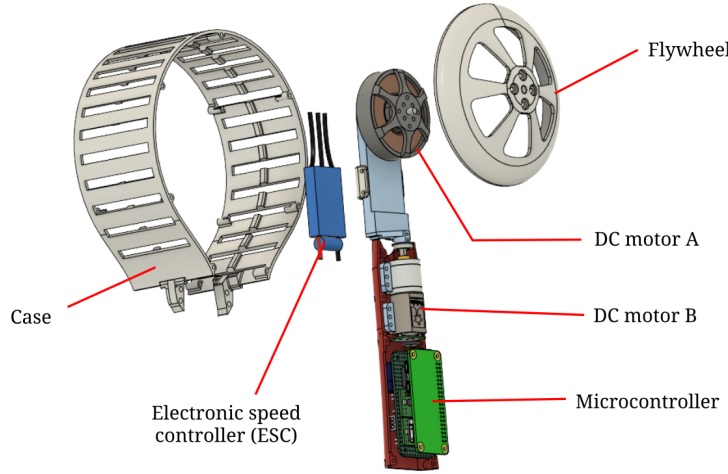


Figure 2: The design and the assembled prototype of our racket device.

of the racket blade, as illustrated in Figure 2. This difference arises from the DC motor B and microcontroller being built inside the handle of the proposed racket.

The proposed racket device was able to generate a gyroscopic force and assist the user's swing motion by using the gyroscopic effect as shown in Figure 3. The gyroscopic effect generates a torque (green arrow) in the yaw axis, perpendicular to the rotation direction of the spinning flywheel (red arrow) in the pitch axis. In Figure 3, by rotating the spinning flywheel clockwise, the device generates a gyroscopic force to assist the user in swinging forward. The device is also capable to produce gyroscopic force against user's swing by rotating the spinning flywheel counter-clockwise.

2.2 Quantitative Evaluation of Gyroscopic Force

We have prepared three different flywheels with a radius $R = 57$ mm and the weights $m = 124$ g, 143 g, and 170 g respectively (including the weight of the DC motor A of 106 g) by modifying the infill and the number of holes in the flywheel. The DC motor A, with an RPM constant of 450 rpm/v, can generate angular velocity $\omega = 9000$ rpm (942.47 rad/s), when 20 V input voltage is applied. The technical specification of the DC motor B describes its angular velocity at maximum efficiency $\dot{\theta} = 120$ rpm (12.56 rad/s) [2]. In theory, the torque output of the gyro moment can be defined as follows:

$$M = \frac{1}{2} m R^2 \omega \dot{\theta} \quad (1)$$

M is the output torque of the gyro moment, m is the weight of the flywheel, R is the radius of the flywheel, ω is the angular velocity of the DC motor A connected with the flywheel (purple arrow as shown in Figure 3), and $\dot{\theta}$ is the angular velocity of the DC motor B to control the rotational direction (red arrow as shown in Figure 3). Therefore, with the current assembly, our proposed racket system can generate a maximum torque of 2.38, 2.74, and 3.26 N·m, respectively for each flywheel's weight. This theoretical value is more than twice the theoretical value from conventional

device proposed by Nakamura et al.[18] with the theoretical value of maximum torque is 1.13 N·m.

To evaluate the force feedback performance of our racket device, we measured the range of presentable gyroscopic force with the experiment setup as shown in Figure 4 (A). We evaluated the gyroscopic force of the device with three flywheels of different weights: 124 g, 143 g, and 170 g. In addition, five different rotation degrees of the flywheel: 5°, 10°, 15°, 20°, and 25° were evaluated. To measure the gyroscopic force, we utilized the IMADA ZTA-1 force measurement device. We fixed the racket to a custom-made rotational board and attached an inelastic cotton string with an approximate length of 30 cm to the backside of a metallic part that connects the DC motor A and the DC motor B as illustrated in Figure 4 (A) and 4 (B). The string was placed as close to the center of the spinning flywheel. We then attached the other end of the string to the force measurement device. The string was kept under tension before each swing to prevent the buildup of kinetic energy. We set the racket device to swing five times for each combination of flywheel and rotation angle. For each swing, the peak value of the gyroscopic force was recorded.

Figure 4 (C) shows the experiment result of the gyroscopic force measurement. The gyroscopic force increased along with an increase in the weight of the flywheel. The gyroscopic force also increased when the rotation angle was increased from 5° to 15°. However, the gyroscopic force did not have any significant change when the rotation angle was increased larger than 15°. As described by Berna et al., we can assume that after the 15° the gyroscopic effect is no longer confined to yaw axis, but also generated in other directions [1].

Our analysis of the gyroscopic force feedback device revealed the discrepancies between theoretical calculations and measured values. The theoretical gyroscopic forces were calculated, as the output torque is the product of force and length of lever arm (distance between center of the spinning flywheel and center of rotation board) which is approximately 14 cm (0.14 m). The theoretical maximum forces for each flywheel weight were calculated as 17.03,

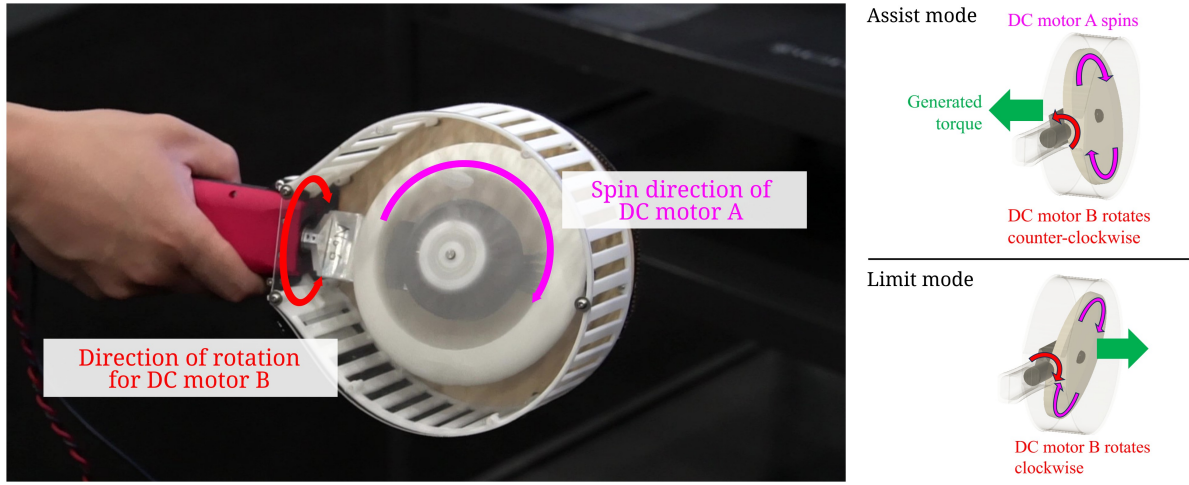


Figure 3: Gyroscopic effect in our proposed racket device. The device can generate torque in yaw axis. The gyroscopic force can be generated in the direction of user's swing and against user's swing.

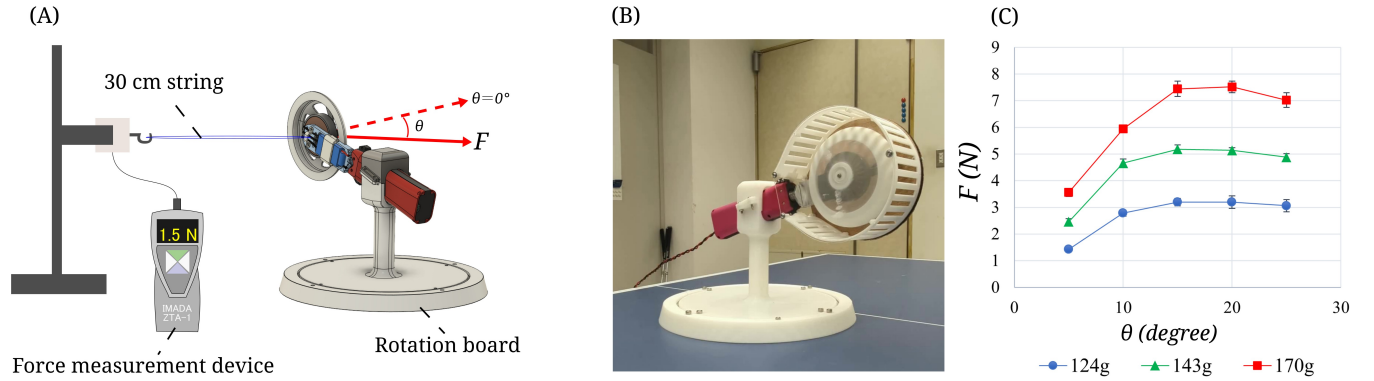


Figure 4: (A) Illustration of the experiment setup. (B) The proposed racket fixed on a rotation board. (C) Experiment result for gyroscopic force measurement.

19.64, and 23.35 N. However, the actual measured maximum forces were 3.20, 5.18, and 7.52 N, respectively. Discrepancies between the measured and theoretical values can be attributed to several factors. Firstly, the theoretical calculations did not account for the motor load, which could have influenced the system's performance. Secondly, the actual angular velocity of the motors may be less than the technical specifications given. Thirdly, frictional forces within the rotation board may have diminished the gyroscopic force, leading to a reduction in the measured values. Despite the discrepancies, our proposed device was able to generate a maximum gyroscopic force of 7.52 N, surpassing the conventional air jet force feedback for racket sports by Tsai et al. [25] and Suzuki et al. [24], which produced gyroscopic force of 3.20 N and 1.50 N, respectively. Our device thus demonstrates more than twice the force of these existing solutions. To switch the direction of torque, we simply change the rotational direction of the DC Motor B. At an angular speed of 12.56 rad/s, the DC motor B can rotate 36° in 50 ms, allowing for a short transition time to switch between the generated torque

directions. With these result, we aim a substantial advancement in force feedback technology for table tennis training.

To measure that the presentable force can return the ball to opponent's area, we utilize newton's second law described as follows:

$$F = m \cdot a \quad (2)$$

The standard mass of the table tennis ball is $m = 2.70$ g. Assuming that the collision between the racket and the ball is elastic, elasticity of the rubber on the paddle is ignored, thus, the both momentum and kinetic energy are preserved [11]. For each flywheel weight with the maximum force of 3.20 N, 5.18 N, and 7.52 N, respectively, the acceleration of the ball after contact a would be approximately 1185 m/s^2 , 1918 m/s^2 , 2785 m/s^2 , respectively. If the ball speed before the racket struck is $v_{\text{initial}} = 2.5 \text{ m/s}$ as simulated by Liu et al. [14] and the duration of force applied to the ball is $t = 0.01$ seconds, the ball speed after being hit by the racket can be obtained by solving the following equation:

$$v_{\text{return}} = -v_{\text{initial}} + a \cdot t \quad (3)$$

The ball's return speed for each flywheel weight are calculated using equation (3), yielding approximately 9.35 m/s, 16.70 m/s, and 25.35 m/s, respectively. Assuming that the user stands 50 cm from the table and holds the racket at a height of 110 cm from the ground, and the user's does not exert any force to swing the racket, we utilized the aerodynamic model (ADM) of table tennis ball introduced by Liu et al. [14] to simulate the ball's movement. For the maximum force produced by each flywheel weight, we found that the returned ball fell 10 cm from the net on the opponent's area with 3.20 N, 10 cm before the edge of the opponent's area with 5.18 N, and out of bounds with 7.52 N. This result indicates that even the user does not exert any force, the maximum force produced by each flywheel weight is sufficient to pull user's hand to return the ball.

2.3 Ball Tracking and Swing Timing Measurement

Ball movement is tracked using four infrared cameras (OptiTrack Flex3, 100 fps): two capturing the opponent's field and two capturing the player's field, as depicted in Figure 5. To predict the optimal hitting moment, the system tracks the ball and predicts its trajectory. We employed ADM developed by Liu et al. [14] to estimate the time required for the ball to reach user's position. After the ball is launched from the machine, the system calculates its position and velocity using data from five consecutive frames. This enables the system to determine the swing timing within approximately 50 ms, allowing sufficient time to provide a swing instruction.

3 Application

3.1 Swing Power Adjustment

The system can also be used to assist or limit the swing power of the player, as illustrated in Figure 6. For example, if a player wants to learn how to hit the ball closer to the net, they need to hit the ball with less swing power. When a player constantly fails to hit over the net with less swing power, the flywheel spins clockwise to generate gyroscopic force in the player's swing direction to assist the swing power. In contrast, it may also help limit a player's swing power. Novice players often attempt to hit the ball as hard as they can [15, 16]. However, excessive power can cause the ball to go out of bounds. To reduce such occurrences, the flywheel spins counter-clockwise to generate gyroscopic force in the opposite direction of the player's swing.

3.2 Bridging The Gap Between Players

Previous studies have introduced the concept of integrated embodiment to form partnership between human-human [21] or human-computer [6, 12, 17]. Such applications have the potential to be integrated in our proposed system to bridge the gap between novice and expert. For example, an expert demonstrates how to swing the ball with the correct amount of force. It is not an easy task for a novice to immediately imitate it after only seeing the demonstration a few times. A computer can sense and analyze the expert's swing power and movement. When the novice swings the racket, the computer communicates with the proposed device to generate a force that complements the novice's swing power to match the expert's swing

power. Additionally, shared agency is also a possible use case. As introduced by Patibanda et al. [21], through the sharing of bodily control, a novice uses their body to react against an incoming ball, while the racket is controlled by an expert to pull the novice's arm. Such an application expands the table tennis playing experience and has the potential to directly teach the novice the appropriate amount of force needed to return the ball.

3.3 Demo System and User Feedback

We created a simple demo system as illustrated in Figure 5. We set the flywheel weight to 143 g and the rotation angle of 15°, allowing the device to generate gyroscopic force up to 5.18 N. We held a public event in our institution where the general public could try our system. Approximately fifty individuals from different backgrounds were able to experience our system. We asked the participants to stand on a tape that we placed approximately 50 cm from the table. Since the proposed device can only assist and limit user's swing power, we asked the participant to freely adjust the racket height. During the demo, all participant confirmed that most of the balls can be returned to the opponent's court, even if the participant doesn't exert any force. From the public event, we received several positive comments as follows: *"It's useful for a beginner. Without having to swing, the racket will force them to swing at the right moment."*, *"I don't have to think or to see the ball to return it. It's interesting."*, *"Being able to return the ball automatically is a mysterious feeling."*, *"The device gives a sense of pulling which made me want to swing."* However, the device still has some room for improvement based on the following comments: *"It's a little bit heavy for a racket."*, *"I need time to learn how to use the device."*

4 Limitation

The current design of the racket is thicker compared to a standard table tennis racket. A standard table tennis racket is about 16 mm thick, while our proposed racket is about 69 mm thick. The case was designed to be thick to contain the motor and accommodate gyroscopic movement. In addition, we received several feedback regarding the weight of the racket during the demo. The current proposed racket weighs approximately 530 g, whereas a standard table tennis weighs around 70 to 100 g [3]. To make the racket sturdy, we incorporated metallic components to connect the DC motor A and the DC motor B (refer to Figure 2). While this design choice enhances stability and durability, it also contributes significantly to the overall weight of the racket. In the future, we consider integrating lighter motors and materials to improve the racket's usability and performance without compromising its structure.

In this study, we focused solely on measuring the gyroscopic force along a single axis. This decision was based on our aim to generate force that produces gyroscopic effect in the forward and backward directions to modify user's swing power. Although we assumed that the gyroscopic force is generated in other directions when the rotation angle exceeds 15° as described by Berna et al. [1], we could not confirm such force, as all users reported feeling the force pulling their arms only in forward or backward directions. In the future, we plan to measure generated torque in other directions and confirm its effect on user's experience.

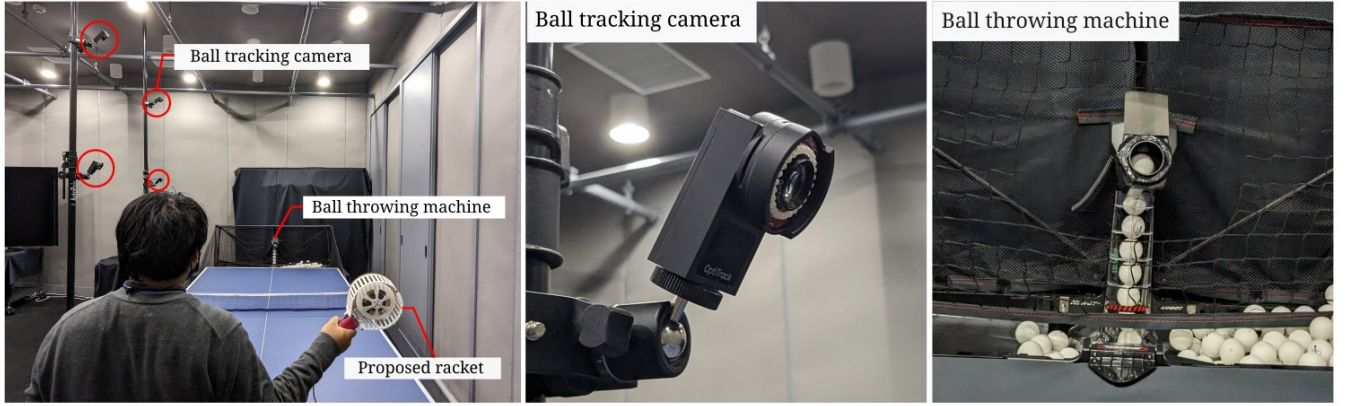
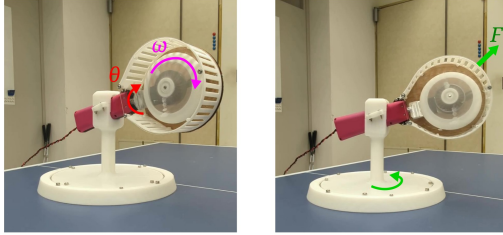


Figure 5: The demo system utilizes ball tracking cameras and a ball throwing machine.

(A) Power assist mode



(B) Power limit mode

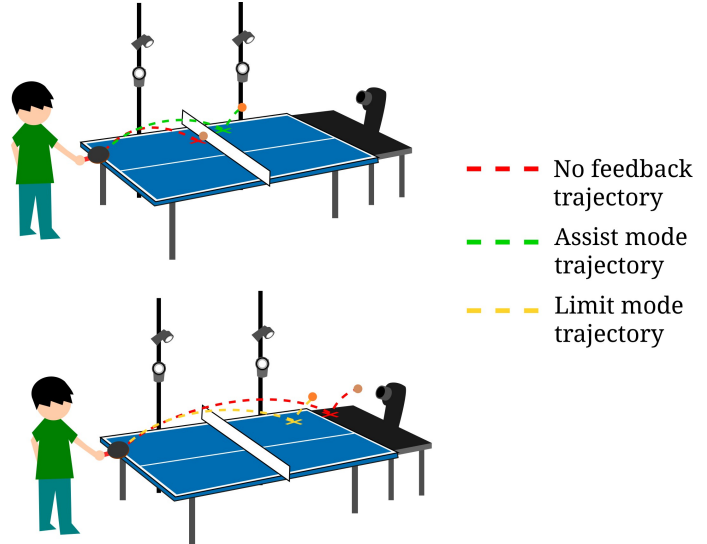
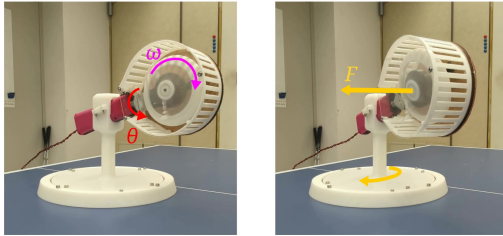


Figure 6: Use case in adjusting player's swing power when player's swing is weak (top) and player's swing is strong (bottom). The direction of rotation of the DC motor B (θ) affects the direction of the generate force to assist or limit user's swing power.

The current demo system can handle a range of ball speeds, but it is limited to responding only to balls traveling in a single direction, effectively addressing only forehand shots. This constraint restricts the system's ability to accommodate more complex motions like backhand shots. Currently, the system can generate gyroscopic force along the yaw axis, enabling it to pull the user's hand forward or push it against user's swing. Although the demo was confirmed with several participants, the current prototype weighs significantly more than a standard paddle, raising concerns about its training effectiveness. However, comments from the demo participants indicate the potential of current device, suggesting that a lighter design may enhance its usefulness for training in the future.

The current case design also includes several narrow holes which could lead to safety concern. The hole in the case was designed as narrow as possible to prevent a child's finger pass through while reducing the total weight of the racket. However, it is still possible

that some small objects may pass through the gap. Therefore, during the demo, we made sure that the workplace was free of debris. In the future, while reducing weight on the main parts, a solid case could be considered.

5 Conclusion

In this study, we proposed ModPaddle, a racket-shaped force feedback device to adjust players' swing power and provide swing timing. Since the proposed device is ungrounded, the user's movement is not restricted while holding the device. The device utilizes the gyroscopic effect to generate a wide range of gyroscopic forces, allowing coaches to instruct the correct amount of power to return the ball successfully. Measurement results suggest that the proposed device can generate forces more than twice as much as that of conventional air jet force feedback devices for racket sports. We demonstrated the application of ModPaddle to assist or limit

the player's swing power in returning the ball to a desired range and discussed the user feedback and limitations for improvement. In future work, we plan to investigate the efficiency of the proposed device in improving the learning rate of swing power control through a user study.

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